LAMMPS-Modeled Radiation Damage in Nuclear Materials

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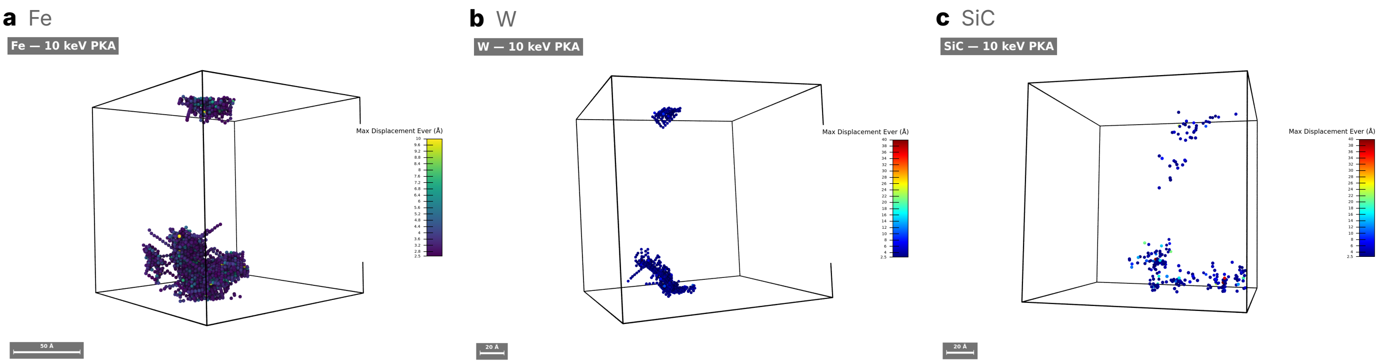

Fig. 1 | Cascade damage at 10 keV in Fe (a), W (b), and SiC (c).

1. What We Did in LAMMPS

Step 1: Built the crystals. 3C-SiC (262,144 atoms), bcc Fe and W (250,000 each), with published potentials[1].

Step 2: Fired the cascade. One central atom given 10 keV at 300 K, aimed away from crystal channels.

Step 3: Counted the damage. Wigner–Seitz.

Step 4: Measured the stored energy. The potential energy left behind after the cell cools back down, the energy locked into the surviving defects.

Step 5: Forward-modelled diffraction. Pair distribution functions G(r), the histogram of all interatomic distances, which is exactly what a diffraction beamline measures, so our damaged cells can be compared directly with synchrotron experiments on irradiated SiC.

2. Results

Fig. 1 shows each crystal after its cascade; every dot is an atom that was knocked out of place. In Fe and W the impact melts one small pocket, and when that pocket refreezes the atoms mostly fall back onto their lattice sites, so W ends up with only 12 lasting defect pairs. In SiC the energy scatters into many small damage pockets instead, because carbon atoms are light and easy to knock loose; 152 defect pairs survive, and the crystal traps about six times more energy than either metal.

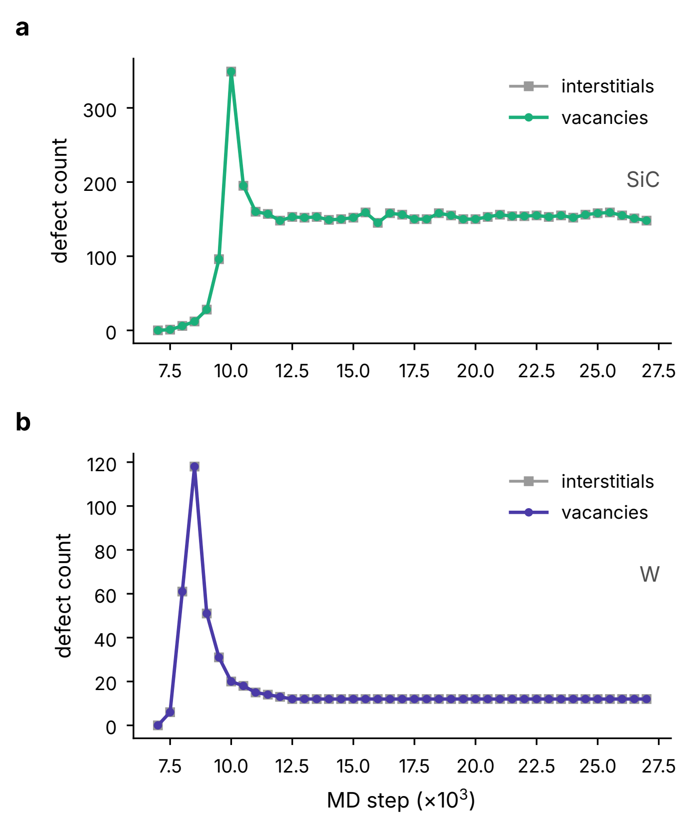

Fig. 2 | Wigner–Seitz defect counts through the spike: SiC (**a**) peaks at 349 and keeps 148; W (**b**) peaks at 118 and keeps 12.

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Table 1 | Key results.

Quantity	This work	Literature	Match
Stored energy, Fe/W/SiC (eV)	122 / 147 / 898	$\leq$1390 J/g at dose[2]	✓
Energy per defect pair, Fe (eV)	4.9	5–6[3]	✓
Energy per defect pair, SiC (eV)	5.7	5–7[1]	✓
Energy per defect pair, W (eV)	12.3	12–13[4]	✓
C share of SiC stored energy	57.9%	C-dominated[5]	✓
C:Si defect population	2:1	35/20 eV thresholds[6]	✓
In-cascade recombination, SiC	58%	~60%[7]	✓
In-cascade recombination, W	90%	~1/3 NRT survival[8]	✓
NRT survival, Fe	0.337	0.364[8]	✓
NRT survival, SiC	1.48	1–2[7]	✓
NRT survival, W	0.337	~0.24[9]	✓
E_d(Si) (eV)	36.2 $\pm$ 2.0	35[6]	✓
E_d(C), soft channel (eV)	20–24	~21[6]	✓
E_d(Fe) (eV)	52.2 $\pm$ 4.1	40[9]	✓
E_d(W) (eV)	102 $\pm$ 10	83–128[10]	✓
W clustered interstitial fraction	0.58	0.5–0.7[11]	✓
Partial-PDF damage, Si-Si vs C-C	equal (66.9)	resolved only at dose[12]	✓
Static disorder ratio, C/Si	1.00	1.25–2.8 with dose[5]	✓

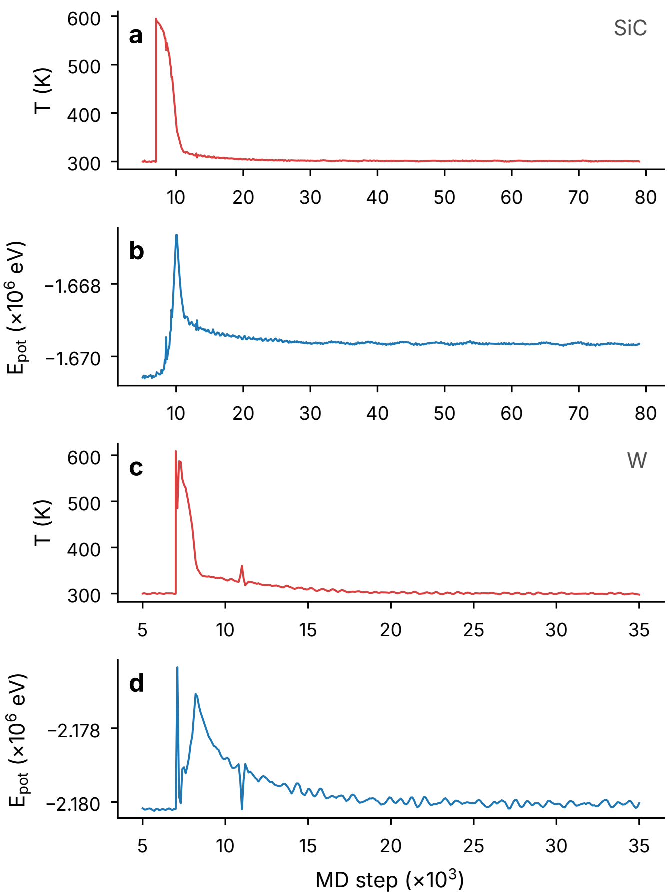

Fig. 3 | Temperature (**a**, **c**) and potential energy (**b**, **d**) for SiC and W; the energy offset that survives cooling is the stored energy.

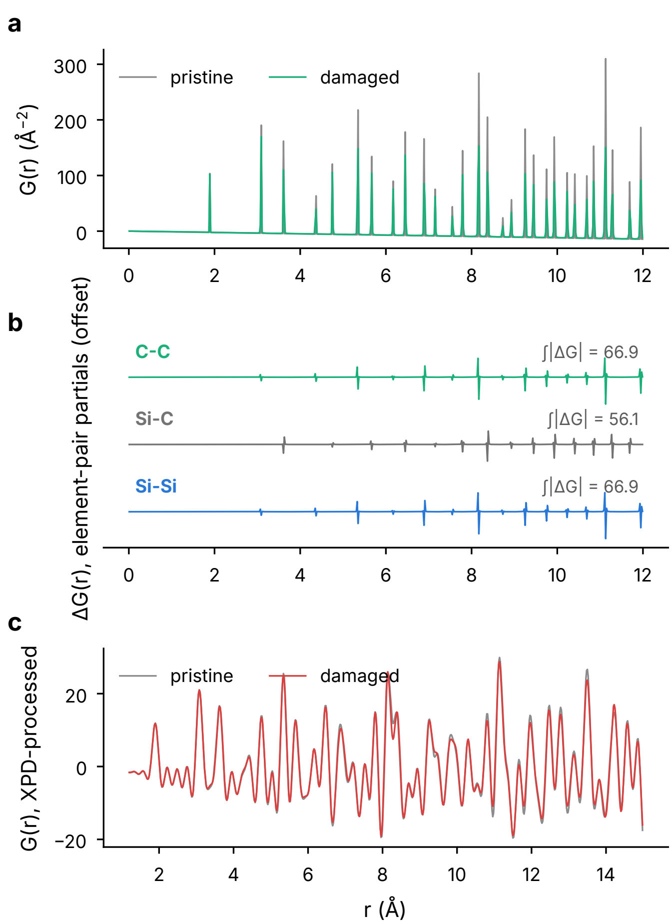

Fig. 4 | Damaged vs pristine G(r) (**a**), its element-pair partials (**b**), and the beamline-processed version (**c**).

3. Visual Heat Strain

Each crystal was also heated in a separate staged NPT test from 300 K to its ramp peak, with every atom colored by how far it has moved from its cold position (Fig. 5), blue near the lattice site, drifting red as thermal vibration and lattice expansion grow. Heating disorders every atom a little and equally, while a cascade (Fig. 1) disorders a few atoms severely, the difference between reversible thermal motion and stored damage.

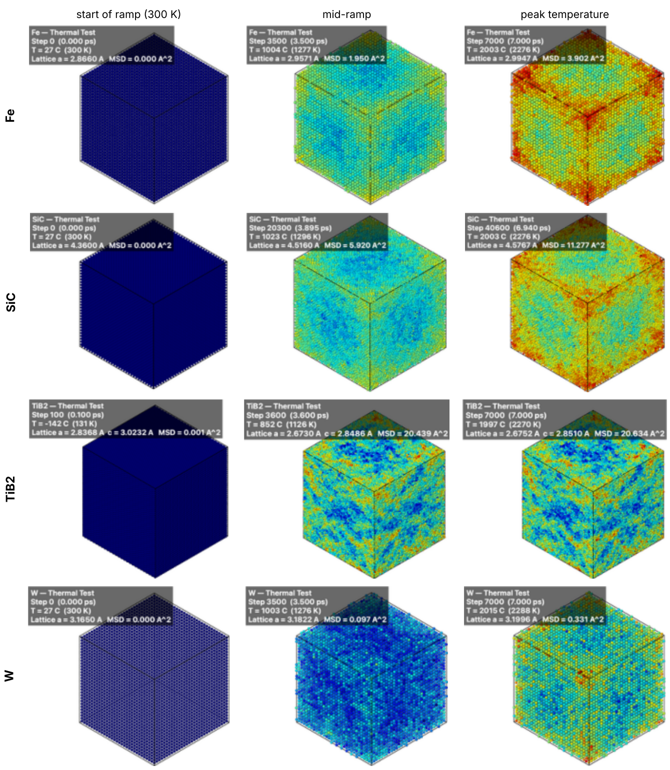

Fig. 5 | Staged heating of Fe, SiC, TiB_2, and W: atom displacement from the cold lattice at the start, middle, and peak of the temperature ramp.

4. References

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